Machine Learning Inference of Cosmic Dawn Parameters from 21-cm Power Spectrum

Joey Wee. MPhil Data Intensive Science, University of Cambridge. Word count: 2997

I. Introduction

THE Cosmic Dawn remains poorly understood. The Square Kilometre Array aims to use measurements of the 21-cm power spectrum to constrain the physical parameters governing the era. Unfortunately, this 21-cm wavelength is dominated by foreground emissions. The weakness of the science signal makes the accurate removal of noise vital.

There is only one observable universe; as such, noise cannot be removed by averaging over independent observations. Instead Bayesian inference is used to fit joint models of noise and science parameters. Bayesian inference compares observed data to forward-modelled theoretical data combining the signal and noise given sets of parameters to infer a posterior over those parameters.

In this coursework, likelihood-based inference and simulation-based inference are used to infer the parameters θ_4, comprising of the universe's expansion rate h, the early galaxy x-ray luminosity $L_{40}^{\text{x-ray}}$, the ionising photons escape fraction f_{esc}^{10}, and the star formation efficiency ϵ. The required simulations will be 21-cm power spectra produced using the Zeus21 cosmic dawn model at k-modes between 0.001 and 0.9 Mpc^{-1} at a fixed redshift $z = 13$ [1].

II. Likelihood-based Inference

Likelihood-based inference uses defined likelihood functions on the observations P^{obs} and the forward-modelled spectra P^{model} to sample a posterior. During inference, P^{obs} must be compared to P^{model} generated for every evaluated θ_4 to compute the likelihood. However, running full astrophysical simulations every likelihood call is computationally intractable as $10^5 \rightarrow 10^6$ likelihood calls may be required. Therefore, the model is replaced with an emulator that can generate P^{model} power spectra: a task for which neural networks are well-suited due to their ability to learn complex relationships in data.

The simulated $P^{\text{model}} \in \mathbb{R}^{54}$, making the output space very high-dimensional for a neural network. Neighbouring samples in the P^{model} spectra are also expected to be correlated. Training an NN to directly output P^{model} would therefore require a model to waste capacity learning these inter-bin dependencies in the spectra, which would be difficult if the loss function treated the outputs as independent.

Instead, the spectra can be decomposed onto basis functions. Since each basis function already encodes the correlated k-structure of neighbouring bins, predicting the coefficients based on an input θ_4 would allow P^{model} to be reconstructed simply with the correlations built into the predictions. This also significantly reduces the dimensionality of the data to predict.

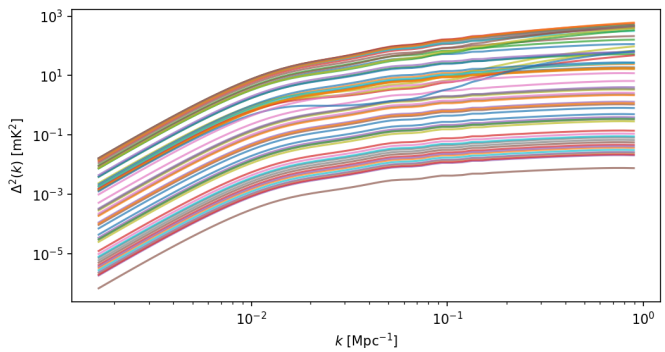

Fig. 1: A set of the P^{model} power spectra plotted log-log space. The spectra vary in magnitude by several orders of magnitude. One example of the spectra exhibits an extra dip.

Finding suitable basis functions to decompose the data was done using PCA.

A. Principal Component Analysis

PCA is a dimensionality reduction technique which fits basis vectors aligned with the directions of greatest variance in a dataset. By retaining only the components capturing the largest fraction of the variance in the data, that data can be simplified significantly with acceptable loss of information.

Examining the simulated power spectra plotted in Fig. 1, the power varies by several orders-of-magnitude. A good PCA should find the most meaningful components, not just the largest magnitude components. Thus, the power spectra are pre-processed by taking the logarithm of the power before performing the PCA; any spectra reconstructed on the resulting PCA components must then be re-exponentiated to yield a comparable real-space signal.

The PCA was performed on a training subset of the observations (a test-train-valid-split of 0.1-0.8-0.1 was used).

The number of components was chosen by examining the explained variance, the fraction of the variance in a dataset that is captured by a subset of the principal components.

Variance threshold	Number of components required
99.00%	1
99.90%	3
99.99%	5

TABLE I: Number of principal components required to explain different fractions of the total variance.

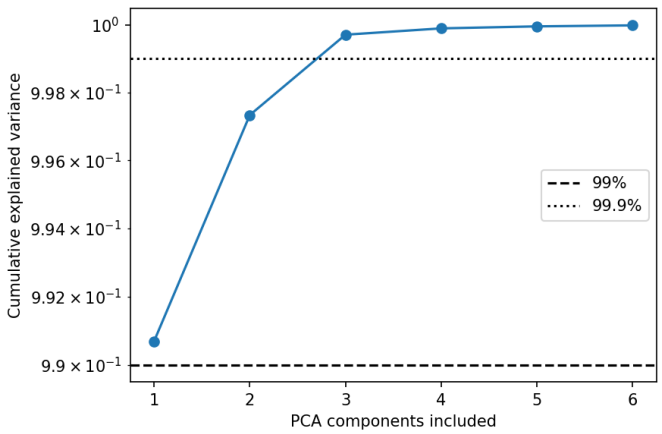

Fig. 2: The cumulative explained variance in the data that is achieved when using different numbers of basis modes created from a PCA on the log-power spectra. The vast majority of the variance is explained by just the first component with reasonable further gains until the third component. After the third component, the marginal increase in explained variance with every extra mode plateaus.

The cumulative explained variance for the PCA components calculated on the set of log-power spectra is plotted in Fig. 2. More than 99% of the variance is explained just by one component, with further gains until the 99.9% threshold at the third component; after this, the explained variance gain plateaus. The number of modes required for cumulative variance thresholds are stated in Table I.

However, the explained variance will not correspond directly to emulator accuracy. Some of the more extreme values of θ for which likelihoods will be required may result in features, such as the occasional spectral dips observed in Fig. 1, which leave low-variance footprints in the PCA due to their rarity. The training set may also not be completely representative and some features that appear more frequently in the validation and test sets may look less important in the training set. Increasing the number of components for safety would not increase compute required too significantly. Therefore, despite more than 99.9% of the variance being explained in 3 components, the likelihood-based inference was conducted using 6 components.

6 component reconstruction performance was examined by plotting examples of reconstructed spectra. One such example is plotted in Fig. 3. The spectrum reconstructed from 6 components agrees well with the sample spectrum, producing small fractional residuals never exceeding the 1% magnitude.

To yield a more complete overview of the training set the mean absolute percentage error (MAPE) of the P^{model} reconstructions was computed and plotted in Fig. 4. The mean MAPE is very small at 0.43% and even in the worst cases at the 95th percentile of error it is only 1.14%. Because the NN will predict the coefficients of these components, small PCA reconstruction error is important as it defines the floor for the accuracy of the NN-predicted spectra.

The overall low reconstruction MAPE indicates that a sufficient components were used. Adding more components would improve this floor performance at the expense of increased compute requirements.

B. Data Normalisation

All the training P^{model} were projected onto these components and the weights will be used as the NN-prediction targets. As can be seen in the left frame of Fig. 3 the coefficients of each mode can vary by an order-of-magnitude. For better training, the weights were normalised using a `Sklearn` StandardScaler to 0 mean and unit variance [2].

The physical parameter distributions, plotted in Fig. 5, were also examined. The unscaled parameters on the top row have significant differences in the scale of the means and order-of-magnitude differences in range, which can significantly impact model training by biasing the input importance towards parameters with larger scale. So, the input parameters were also scaled using.

It is notable that the distributions of f_{esc}^{10} and ϵ look very similar, with the same mean and range. This could indicate some correlation between these parameters.

C. Likelihood-based Inference Model Training

A fully-connected MLP was chosen to predict the 6 coefficients based on the $\theta \in \mathbb{R}^4$ to maximise model flexibility. For best performance, the hyperparameters were tuned using Optuna [3]. The search varied depth and width of the model, as well as the learning rate and weight decay regularisation to discourage overfitting. The search domain is enumerated in Table II.

Hyperparameter	Type	Search Space	Best Value
Layer Count	Categorical	$\{n\}_{n=3}^{10}$	3
Hidden dimension	Categorical	$\{2^n\}_{n=5}^{9}$	128
Learning rate	Log-uniform	$[10^{-5}, 10^{-2}]$	4.86×10^{-3}
Weight decay	Log-uniform	$[10^{-6}, 10^{-2}]$	1.05×10^{-5}

TABLE II: Optuna hyperparameter search space and best-performing model for minimising the MSE loss on the log-power-spectrum predictions. The best trial selected a model with 3 layers, hidden dimension 128, learning rate 4.86×10^{-3}, and weight decay 1.05×10^{-5}.

MSE reconstruction error against the training power spectra in log-power-spectrum space was used as the loss for physical interpretability of the results. The best model parameters found using this loss by the Optuna search are displayed in Table II.

Early stopping was used to select the final model based on the best validation loss to maximise model generalisation.

NN performance was assessed by plotting a subset of predicted reconstructions against the truths. This indicates good model predictive performance as the model is able to follow the spectra well, predicting even the fine scale structure.

A broader view of the predicted reconstruction accuracy can be obtained by plotting the distribution of the mean MAPE of

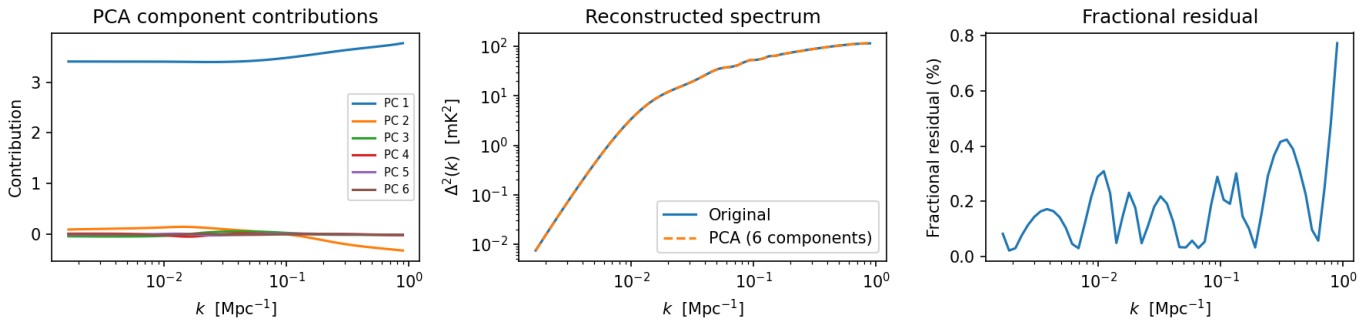

Fig. 3: An example spectrum reconstructed using just the first 6 principal components. Left: the principal components from which the spectrum was reconstructed. Centre: the reconstructed spectrum plotted against the full spectrum. The spectra agree well. Right: The fractional residuals across each k-bin remain below 1%.

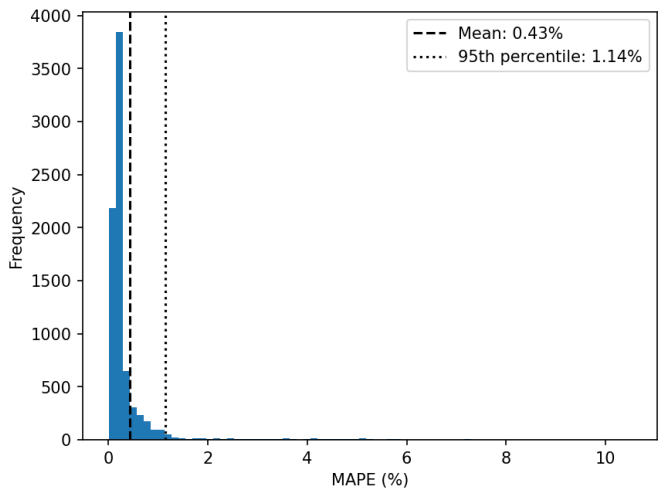

Fig. 4: The MAPE of the PCA-reconstruction power spectra against the full spectra. The MAPE mean is of sub-percent scale and the 95th percentile error is barely above 1%.

the reconstructions (Fig. 7). The mean MAPE on the model-predicted reconstructed spectra is now 1.65%, which is still on the order of 1% and is not significantly higher than the model performance floor due to PCA reconstruction error. However, this error distribution is quite long-tailed with a 95th percentile error at 5.63%, which could become problematic during posterior sampling, since a non-trivial fraction of likelihood evaluations would be based on inaccurate emulator predictions.

Model performance could be improved by using more PCA components and running a larger hyperparameter optimisation. Such a model was trained for this coursework but the inference time on that model made the posterior sampling prohibitively slow.

D. Likelihood-based Inference Posterior Sampling

Now using the model emulator, the posterior can be sampled. Assuming that the theoretical model is perfect and that any deviations are the result of Gaussian noise fluctuations, a likelihood can be defined as

$$\ln L = -\frac{1}{2}\sum_k \frac{\left(P^{\text{obs}}(k) - P^{\text{model}}(k,\theta)\right)^2}{\sigma^2(k)} - \frac{1}{2}\sum_k \log\left(2\pi\sigma^2(k)\right),$$

where the variance is given as

$$\sigma^2(k) = \frac{2\left(P^{\text{model}}(k) - P^{\text{noise}}(k)\right)^2}{N_k},$$

with noise power

$$P^{\text{noise}}(k) = f^{\text{noise}} P^{\text{model}}.$$

f^{noise} is a nuisance parameter with domain $f^{\text{noise}} \in [10^{-3}, 10^{0}]$. N_k is assumed to be 100.

An analytic likelihood like this is dependent on the assumption that all instrumental systematics and non-Gaussian noise have been removed. It also assumes that the k-bins are independent, which is an oversimplification given that the spectrum comes from the same physical processes.

Because the domain of θ_4 varies by orders-of-magnitude between components, it is better to sample in log-space to regularise the shape of the distribution, allowing for better coverage.

There is no prior knowledge of the parameters' final values so an ignorance prior was chosen to be uniform across θ_4 space; due to the log-space sampling a log-uniform distribution was defined to conserve probability mass. Support bounds for the log-uniform prior were defined by finding the maximum and minimum values of all the training parameters and multiplying by 1.5 and 0.5 to yield a buffer. The given range for nuisance parameter f^{noise} was used.

The logarithm of the log-uniform prior is used to find the posterior density by summing with the log-likelihood.

$$\ln P = \ln \mathcal{L} + \ln \pi$$

The ensemble sampler from `emcee` was used [4]. This uses an ensemble of walkers to define the proposal distribution based on the positions of other walkers, allowing it to vary based on the scale of the posterior in each direction so the proposal distribution does not have to be manually tuned.

The MCMC yielded a respectable 0.299 acceptance fraction but a high mean autocorrelation time of 343.86 steps, indicating that the chain is not exploring the posterior efficiently.

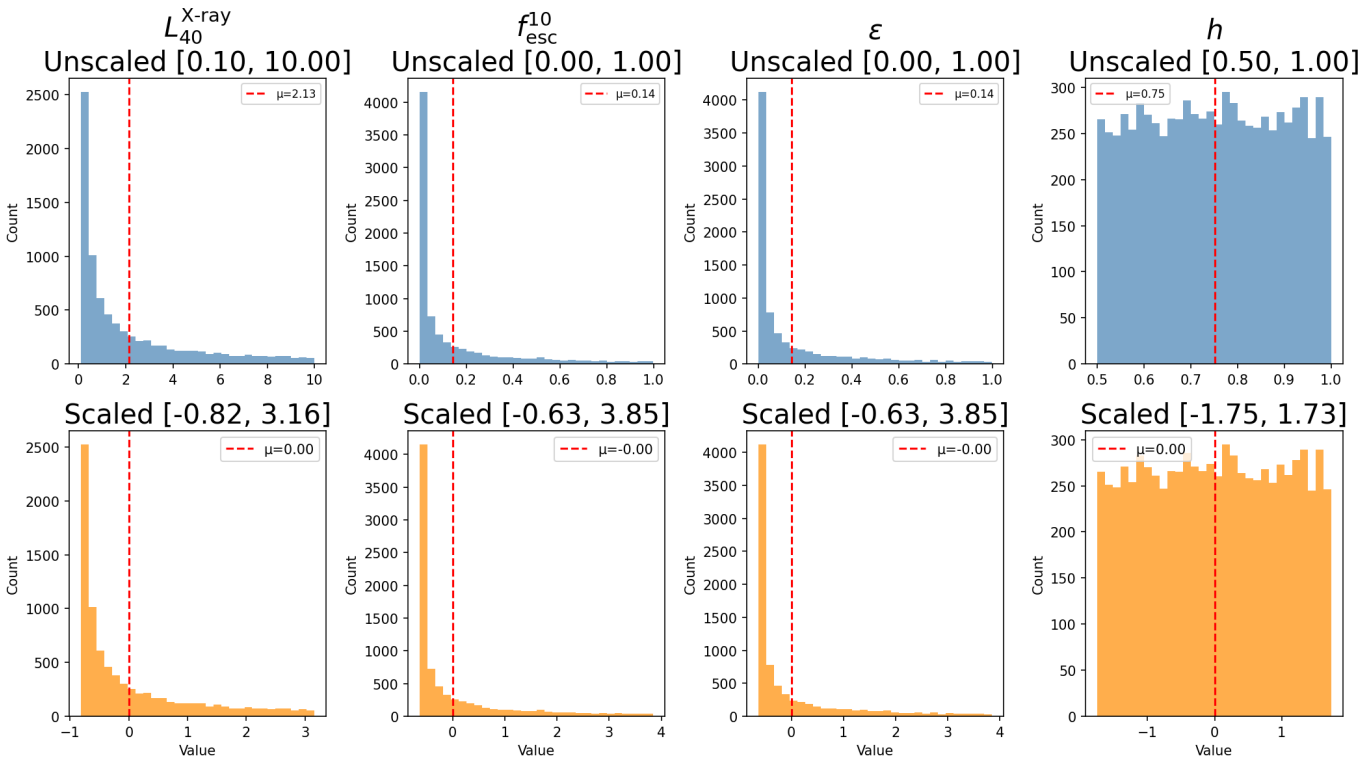

Fig. 5: The distributions of the four physical training parameters before scaling (blue) and after scaling (yellow). The parameter ranges span several orders-of-magnitude, motivating scaling. The scaled parameters all vary in order unity.

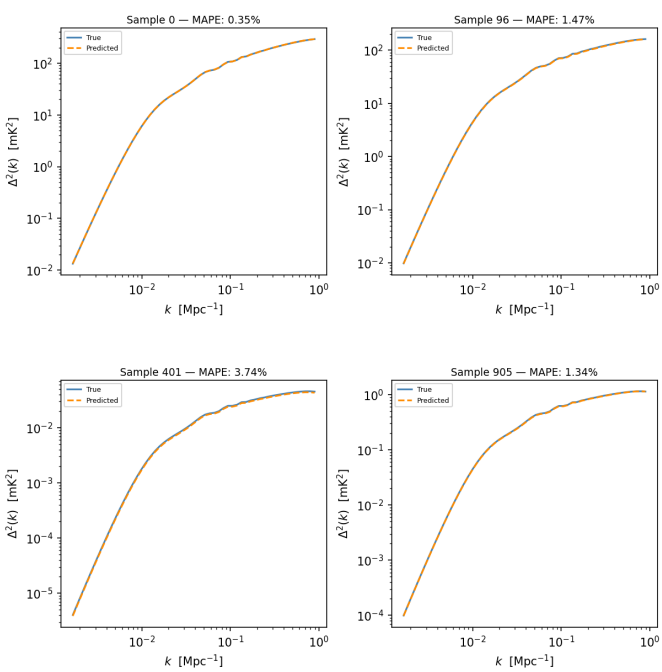

Fig. 6: A random subset of power spectra and their predicted reconstructions. The predicted spectra agree well with the truths. The model is able to follow fine-scale variations.

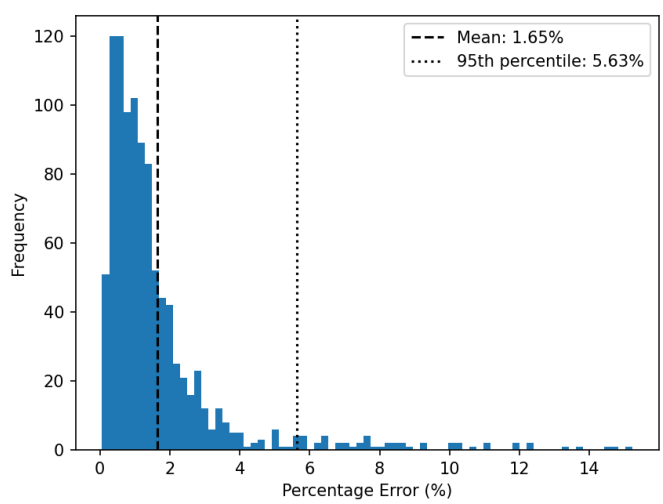

Fig. 7: The MAPE of the NN-predicted reconstructed power spectra against the full spectra. The MAPE mean is just above percent scale at 1.65%. This distribution has a long tail with a 95th percentile error at 5.63%.

One possibility is if the parameters are degenerate or highly correlated, which would warp the high-likelihood region to be very thin and hard to sample.

Sampling quality was examined by plotting the corner plot of the MCMC posterior in Fig. 8. f^{noise}, L_{40}^{xray}, and f_{esc}^{10} have collapsed to prior boundaries. There are no clear posterior peaks: there are entire bands of space with high posterior

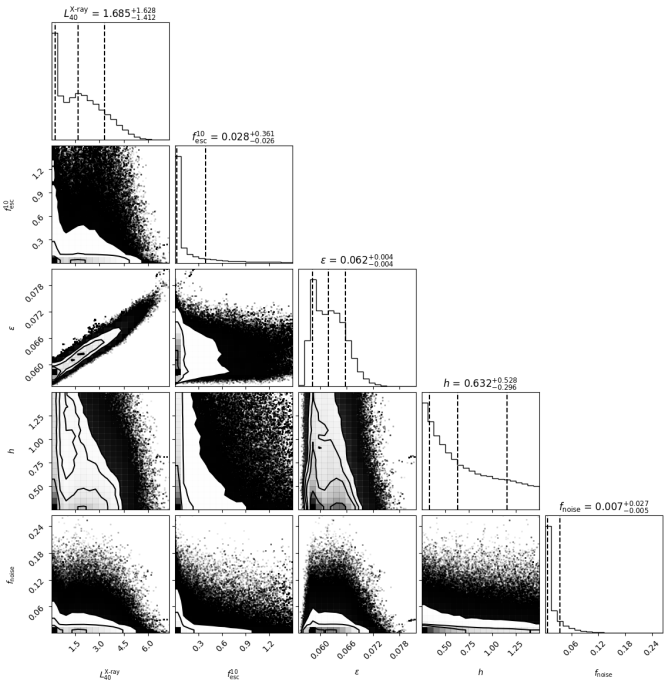

Fig. 8: Posterior sampled using `emcee` ensemble MCMC over a log-uniform prior and a likelihood defined by the emulator. f^{noise}, L_{40}^{xray}, and f_{esc}^{10} have collapsed to prior boundaries. This results in highly poorly defined posterior peaks with large bands of high density.

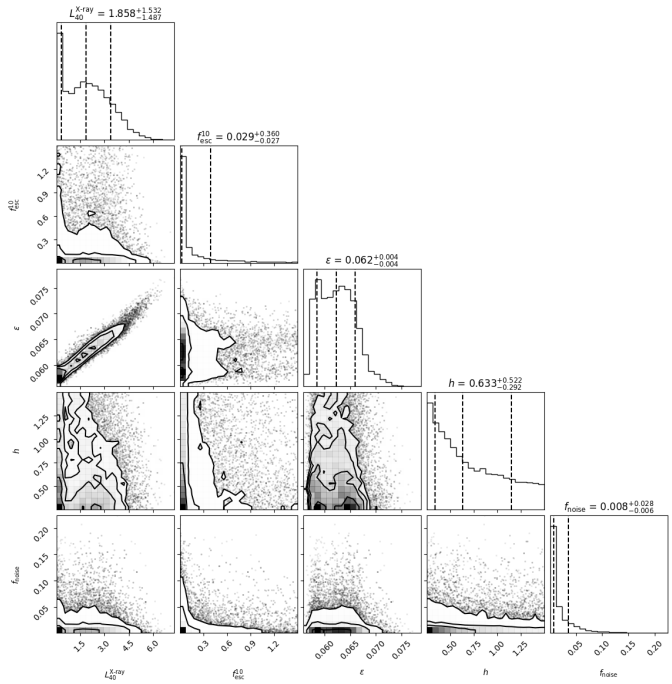

Fig. 9: Posterior sampled using `dynesty` nested sampling over a log-uniform prior and a likelihood defined by the emulator. f^{noise}, L_{40}^{xray}, and f_{esc}^{10} have collapsed to prior boundaries. This results in highly poorly defined posterior peaks with large bands of high density.

density.

Due to the poor MCMC result, the posterior was resampled using a dynesty nested sampling for comparison [5]. The nested sampling posterior across the same log-uniform prior and analytic likelihood is plotted in Fig. 9.

The nested sampler returns the same contours as the MCMC sampler. The median and 1σ credible intervals for each parameter are elucidated in Table III. All parameter constraints are consistent within the very large uncertainties between samplers. This strongly implies that the poor sampling is not due to an error in the sampler configuration but is the result of an unconstrained posterior. The fact that there is only a single observation and a relatively large parameter space supports this interpretation; more samples taken at different redshifts could further constrain these parameters. It is also possible that the likelihood may be too crude to describe the data. We can test this likelihood hypothesis using likelihood-free simulation-based inference.

III. Simulation-based Inference

Simulation-based inference infers a posterior without an explicit likelihood definition using a large ensemble of simulated data produced. If the noise contribution is marginalised over, it is reasonable to expect that a set of parameters θ_4 is informative of the true parameters when the corresponding synthetic spectrum P^{model} is similar to the observed spectrum P^{obs}. For this coursework, a form of SBI called a neural ratio estimator (NRE) is used.

A. Neural Ratio Estimator

NREs are trained on two sets of data: the joint distribution (label = 1)

$$(\theta, x) \sim p(x \mid \theta)\, p(\theta),$$

and the disjoint (label = 0)

$$(\theta, x) \sim p(x)\, p(\theta).$$

The model is trained to distinguish samples drawn from the distributions, thus learning if data came from a set of parameters.

Examine the ratio

$$r(x, \theta) = \frac{p(x, \theta)}{p(x)\, p(\theta)}.$$

Using Bayes' theorem,

$$r(x, \theta) = \frac{p(\theta \mid x)\, p(x)}{p(x)\, p(\theta)} = \frac{p(\theta \mid x)}{p(\theta)}.$$

Thus, the ratio is simply the posterior divided by the prior,

$$r(x, \theta) = \frac{p(\theta \mid x)}{\pi(\theta)}.$$

Taking the logarithm and rearranging gives

$$\ln p(\theta \mid x) = \ln r(x, \theta) + \ln \pi(\theta),$$

which we can use to sample the posterior.

Mathematically, the ideal binary cross-entropy (BCE) classifier output is equivalent to learning r. Removing the sigmoid from the classifier means the model returns $\ln r(x, \theta)$ directly.

Parameter	Emcee Ensemble MCMC	Dynesty Nested Sampling
$L_{40}^{\rm xray}$	$1.69^{+1.63}_{-1.41}$	$1.76^{+1.54}_{-1.44}$
$f_{\rm esc}^{10}$	$0.03^{+0.36}_{-0.03}$	$0.03^{+0.36}_{-0.02}$
ϵ	$0.06^{+0.00}_{-0.00}$	$0.06^{+0.00}_{-0.00}$
h	$0.63^{+0.53}_{-0.30}$	$0.63^{+0.53}_{-0.30}$
$f^{\rm noise}$	$0.01^{+0.03}_{-0.01}$	$0.01^{+0.03}_{-0.01}$
Mean posterior structure	Broad, poorly constrained	Broad, poorly constrained
Boundary collapse	$f_{\rm esc}^{10}$, $f^{\rm noise}$, $L_{40}^{\rm xray}$	$f_{\rm esc}^{10}$, $f^{\rm noise}$, $L_{40}^{\rm xray}$

TABLE III: Comparison between the posterior summaries obtained using `emcee` ensemble MCMC and `dynesty` nested sampling. Both methods yield very similar posterior constraints but with large uncertainties. The parameters $f_{\rm esc}^{10}$, $f^{\rm noise}$, and $L_{40}^{\rm xray}$ remain poorly constrained and collapse towards prior boundaries, while ϵ is the only parameter that is tightly constrained.

B. Generating Noisy Data

To learn the likelihood, NREs need to see instrument noise and systematics. The training vectors must now also include the noise parameter $f^{\rm noise}$: θ_5. Assuming again that the noise model is Gaussian as implied by the emulator likelihood, $f^{\rm noise}$ was drawn from a log-uniform distribution across the supported domain to reflect ignorance of the values to define the variances as above. The simulated model values were taken to be the means.

$$P^{\rm noisy}(k) \sim \mathcal{N}\left(P^{\rm model}(k), \frac{2\left(P^{\rm model}(k) - P^{\rm noise}(k)\right)^2}{N_k}\right)$$

Several $f^{\rm noise}$ can be drawn for each model. The greater the number, the better accuracy we can expect from the model as the training set will cover a wider domain. A larger training set also increases the compute resources needed for training. A compromise of 10 $f^{\rm noise}$ per $P^{\rm model}$ was chosen.

The noisy dataset was verified by plotting them with the base spectra. This is seen in Fig. 10. It is worth noting that the noise is purely Gaussian here whereas real noise would be somewhat correlated between points.

C. Creating Joint and Disjoint Datasets

Using the θ_5 and their corresponding spectra, joint and disjoint datasets were created. To ensure all the data was retained, each set was used twice. Once, preserving the ordering of θ_5 and once shuffling the θ_5. These were concatenated with the corresponding labels of 0 and 1. The overall datasets were shuffled again to mix the joint and disjoint values inside.

To verify that this process creates distinguishable datasets the classifier can learn, samples from the joint and disjoint distributions were plotted in the $(\epsilon, P_{\rm noisy}(k = 0.1))$ plane. The joint samples show a clear correlation between ϵ and $P_{\rm noisy}$ but this relationship is not preserved by the disjoint distribution. Since the distributions are clearly different, the model can learn to classify between them.

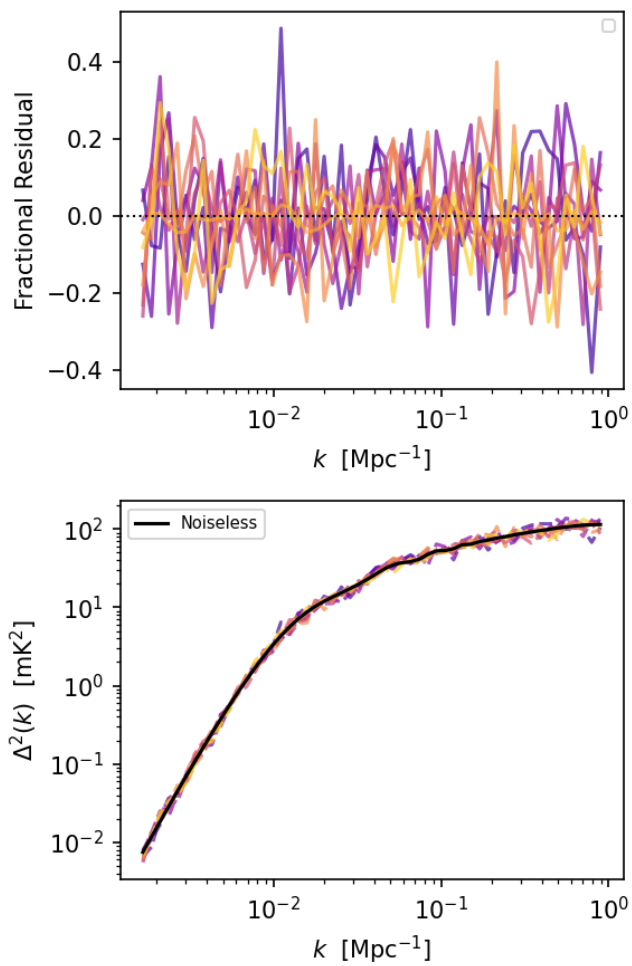

Fig. 10: An example of the noisy samples drawn from a single simulated spectrum. Top: the Gaussian noise "residuals" drawn for 10 draws of $f^{\rm noise}$. Bottom: the residuals overlaid onto the simulated spectrum.

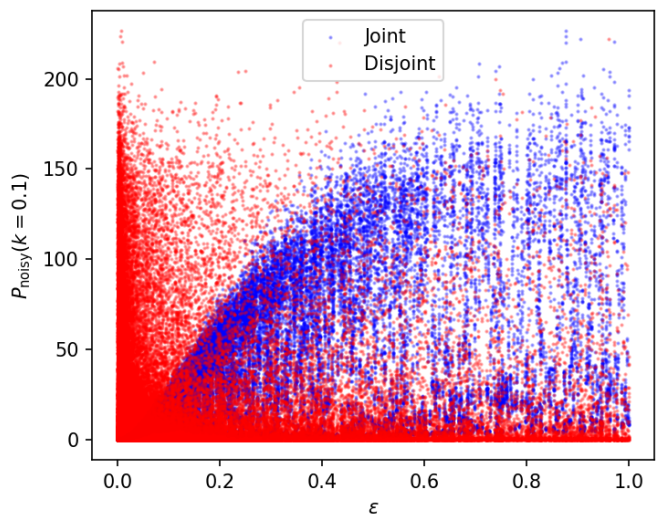

Fig. 11: Samples from the joint and disjoint distributions plotted in the $(\epsilon, P_{\rm noisy}(k=0.1))$ plane. The joint samples (blue) show a clear correlation between ϵ and $P_{\rm noisy}$. The disjoint samples (red) do not preserve this relationship, illustrating that the two distributions are distinguishable.

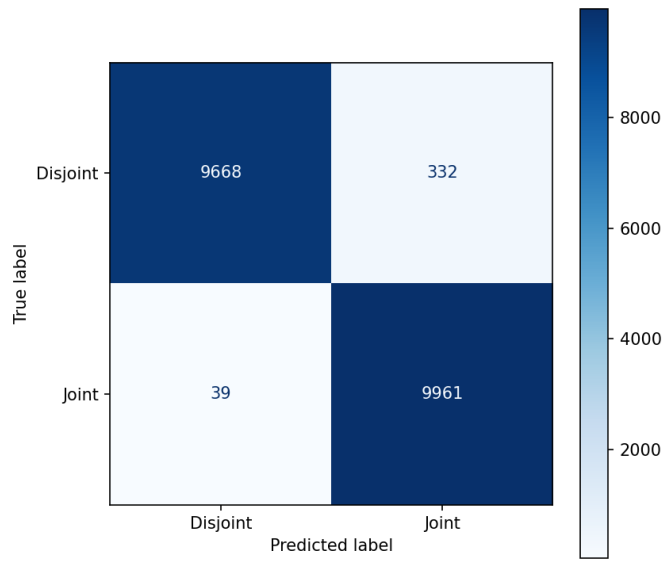

Fig. 12: The confusion matrix of the NRE classifier over the test set. Most values fall along the correctness diagonal but a non-trivial portion of the disjoint samples are labelled by the model as joint.

D. NRE Training

Because multiple $f^{\rm noise}$ were drawn for each sample and the datasets were duplicated during the creation of the joint-disjoint datasets, the datasets for training the NRE were 20 times longer than the emulator datasets. Due to the significantly larger number of training samples, a simpler dense model was used without Optuna tuning to maintain tractability. This model contained an input dimension of 59, corresponding to the concatenation of $\theta \in \mathbb{R}^5$ and $P^{\rm noisy} \in \mathbb{R}^{54}$. The input fed two hidden layers of width 128 and ReLU in between. The output layer was linear so that the network returned $\ln r(x, \theta)$ directly, while the sigmoid was incorporated implicitly through the binary cross-entropy-with-logits loss.

The model was selected by the best validation loss to get the best generalising model. This model's binary-cross-entropy test loss was 0.0738, which is very low, indicating good classification performance. The confusion matrix, plotted in Fig. 12, shows that most of the datapoints are classified correctly, suggesting the model has learned a strong representation of the joint distribution. However, a non-trivial amount of disjoint distributions are labelled as joint, which introduces noise into the posterior. A more complete study should run a full hyperparameter tuning to improve the classification performance.

It is difficult to transfer such metrics into anything indicating real-world performance as with the predicted spectra so the performance had to be assessed by sampling the posterior.

E. NRE Sampling

For repeatability between the LBI and the NRE, the same log-uniform prior was used. An `emcee` ensemble MCMC was used to sample from the posterior in log-space to normalise the shape of the distribution. The sampling yielded a low acceptance fraction of only 0.235 and a huge autocorrelation time of 1077.1 steps. Such a large autocorrelation time again implies that the parameters could be degenerate or correlated. The sampling had to be run with 1 million steps to yield chains with decent posterior coverage. These are plotted in Fig. 13.

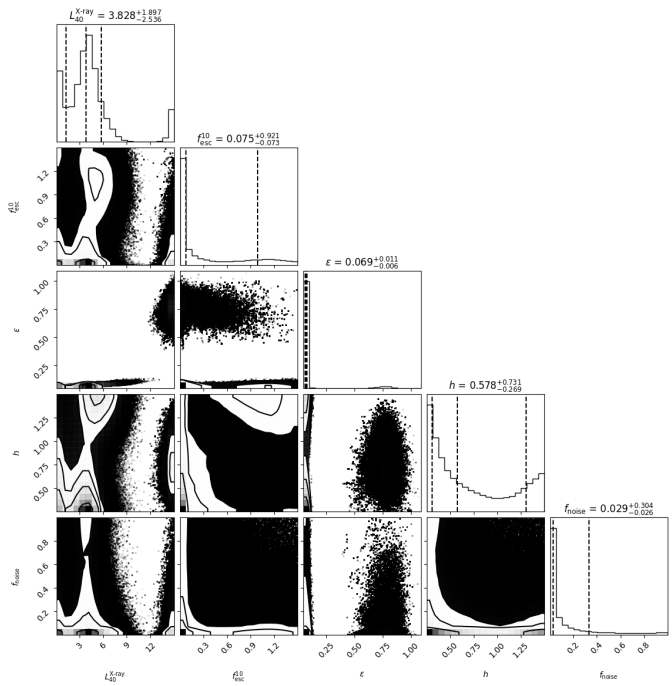

Fig. 13: NRE posterior sampled using `emcee` ensemble MCMC over a log-uniform prior. $f^{\rm noise}$, ϵ, and $f_{\rm esc}^{10}$ have collapsed to prior boundaries. This results in highly poorly defined posterior peaks with large bands of high density.

Similarly with the emulator-based posterior, this MCMC

Parameter	LBI Emcee	LBI Dynesty	NRE Emcee	NRE Dynesty
L_{40}^{xray}	$1.685^{+1.628}_{-1.412}$	$1.858^{+1.532}_{-1.487}$	$3.828^{+1.897}_{-2.536}$	$2.350^{+1.833}_{-2.054}$
f_{esc}^{10}	$0.028^{+0.361}_{-0.026}$	$0.029^{+0.360}_{-0.027}$	$0.075^{+0.921}_{-0.073}$	$0.804^{+0.610}_{-0.800}$
ϵ	$0.062^{+0.004}_{-0.004}$	$0.062^{+0.004}_{-0.004}$	$0.069^{+0.011}_{-0.006}$	$0.062^{+0.005}_{-0.006}$
h	$0.632^{+0.528}_{-0.296}$	$0.633^{+0.522}_{-0.292}$	$0.578^{+0.731}_{-0.269}$	$0.684^{+0.595}_{-0.350}$
f^{noise}	$0.007^{+0.027}_{-0.005}$	$0.008^{+0.028}_{-0.006}$	$0.029^{+0.304}_{-0.026}$	$0.040^{+0.564}_{-0.037}$
Boundary collapse	f_{esc}^{10}, f^{noise}, L_{40}^{xray}	f_{esc}^{10}, f^{noise}, L_{40}^{xray}	f_{esc}^{10}, f^{noise}, ϵ	f_{esc}^{10}, f^{noise}, L_{40}^{xray}

TABLE IV: Comparison between the posterior summaries obtained using the likelihood-based inference (LBI) emulator likelihood and the neural ratio estimator (NRE), each sampled with both `emcee` ensemble MCMC and `dynesty` nested sampling. All sampling methods have samples which have collapsed to the prior boundaries.

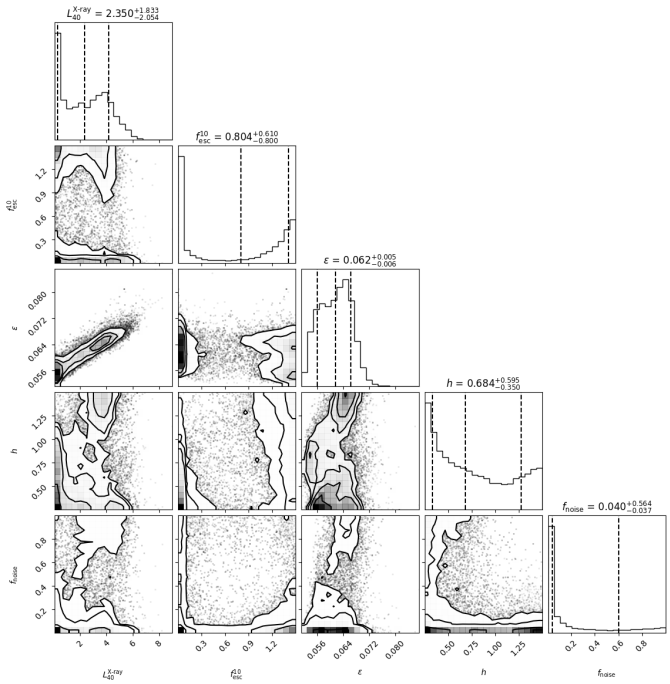

Fig. 14: NRE posterior sampled using nested sampling over a log-uniform prior. f^{noise}, L_{40}^{xray}, and f_{esc}^{10} have collapsed to prior boundaries. This results in highly poorly defined posterior peaks with large bands of high density.

posterior did not converge: there are bands of high posterior density and many of the parameters have collapsed to the prior boundary. The NRE MCMC posterior distribution differs from the LBI ones. To check if the sampling was the problem, nested sampling was also applied to the NRE posterior. The posterior is plotted in Fig. 14. The difference between the two posteriors using the NRE method is quite large; the nested sampling posterior looks more like to the LBI posteriors. Since MCMC showed strong autocorrelation, the nested-sampling result is the more trustworthy posterior in this case. The difference further implies that the MCMC chain has not adequately explored the full NRE posterior.

The summary statistics of all four sampled posteriors are provided in Table IV. Three of the four sampling methods agree qualitatively. The NRE MCMC can discarded because it clearly has not converged. Numerically, the three converged posteriors have some variation between their means and 1σ uncertainties. However, this is because different shares of samples have collapsed to each prior boundary, shifting the numerical values. This suggests that there is some degeneracy in the parameters and the sampler is pulling parameters to extremes to compensate for one another. The only tightly constrained parameter across all converged posteriors was ϵ, suggesting that the 21cm power spectrum at $z = 13$ is most sensitive to the star formation efficiency.

IV. Conclusion

In this coursework, two Bayesian inference approaches were implemented to infer the posterior on an observation of the 21cm power spectrum using Zeus21 simulations.

For each of the LBI and NRE SBI methods, nested and MCMC sampling were implemented. The NRE MCMC was disregarded due to non-convergence and the remaining three methods produced qualitatively similar posteriors, with broad posteriors and considerable density at the prior boundaries. The qualitative agreement between the NRE and LBI posteriors is consistent with that both models assume the same Gaussian noise distribution.

This agreement suggests that the observed behaviour is more likely caused by weakly constrained parameters under the current data and modelling assumptions, rather than sampler failure. This limitation could come from degenerate parameters or insufficient information as only one observation at one redshift was used. Since the physics of how each parameter influences the 21cm signal is different, a single power spectrum cannot easily discern their individual contributions.

Further studies can be performed on these data to confirm that these are the problems. While the NRE test loss is already low, it remains to be seen if a better classifier would improve the sampling. An Optuna search can be performed over the NRE model: if the posterior inference does not improve for a better classifier, that would suggest that the posterior geometry is the problem rather than the classifier. It could also be further quantified that the LBI and NRE are sampling from

the same posterior landscape. Samples of θ drawn from the NRE posterior could be input into the LBI emulator and the resulting spectra compared with the simulated spectra. Agreement between the inferred distributions could then be quantified by the KL-divergence between them.

REFERENCES

[1] Julian B. Muñoz. "An Effective Model for the Cosmic-Dawn 21-cm Signal". In: *Monthly Notices of the Royal Astronomical Society* (2023). DOI: 10.1093/mnras/stad1512. arXiv: 2302.08506 [astro-ph.CO].

[2] Fabian Pedregosa et al. "Scikit-learn: Machine Learning in Python". In: *Journal of Machine Learning Research* 12 (2011), pp. 2825–2830.

[3] Takuya Akiba et al. "Optuna: A Next-generation Hyperparameter Optimization Framework". In: *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. 2019, pp. 2623–2631. DOI: 10.1145/3292500.3330701.

[4] Daniel Foreman-Mackey et al. "emcee v3: A Python Ensemble Sampling Toolkit for Affine-Invariant MCMC". In: *Journal of Open Source Software* 4.43 (2019), p. 1864. DOI: 10.21105/joss.01864.

[5] Joshua S. Speagle. "dynesty: A Dynamic Nested Sampling Package for Estimating Bayesian Posteriors and Evidences". In: *Monthly Notices of the Royal Astronomical Society* 493.3 (2020), pp. 3132–3158. DOI: 10.1093/mnras/staa278. arXiv: 1904.02180 [astro-ph.IM].